

2008 Ford Edge SE

2008 TRANSMISSIONS Rear Drive Halfshafts - Edge & MKX

2008 TRANSMISSIONS

Rear Drive Halfshafts - Edge & MKX

SPECIFICATIONS

GENERAL SPECIFICATIONS

Item	Specification
Left rear halfshaft assembled length	863.9 mm (34.01 in)
Right rear halfshaft assembled length	915.6 mm (36.05 in)

LUBRICANTS AND CAPACITIES

Lubricant	Specification	Capacity
Rear Halfshaft Inboard Constant Velocity (CV) Joint		
Constant Velocity Joint Grease supplied in boot kit 4L537	-	108 g (3.80 oz)
Rear Halfshaft Outboard CV Joint		
Constant Velocity Joint Grease supplied in boot kit 4L537	-	88 g (3.10 oz)

TORQUE SPECIFICATIONS

Description	Nm	lb-ft
Brake caliper hose bracket bolt	15	11
Lower arm outboard bolt	150	111
Lower shock absorber nut	80	59
Rear wheel hub nut (inner)	275	203
Rear wheel hub nut (outer)	175	129
Stabilizer bar link nut	35	26
Toe link bracket bolt	200	148
Toe link outboard bolt	150	111
Upper arm outboard bolt	150	111

DESCRIPTION AND OPERATION

REAR DRIVE HALFSHAFTS

Rear Drive Halfshafts

The rear drive halfshafts consist of the following:

- Inner constant velocity (CV) joints
- Outer CV joints
- Interconnecting shafts

- Convoluted CV joint boots
- CV joint boot clamps
- Special CV high-temperature grease
- Tripod joint housings
- Ball and cage housings
- Circlip stopper that holds the cross-groove inboard race assembly (inboard CV joint) together
- Circlip that retains the splined inboard CV joint to the differential side gear
- RH and LH halfshafts that are different lengths, with the LH halfshaft being the longer of the two

The rear drive halfshafts link the rear axle to the rear wheel hubs. The CV joints allow the halfshafts to rotate smoothly through the required changes in angles between the stationary axle and the varying angles of the rear wheels including jounce and rebound. The CV joints are either tri-lobed style or the caged-ball type, depending on inboard or outboard. New halfshaft components are available in kit 4L537 which contains grease, circlips, boots and clamps.

Handle all halfshaft components carefully during removal and installation and during various component disassembly and assembly procedures. Never pick up or hold the halfshaft only by the inboard or outboard CV joint. Do not overangle the CV joints. Damage will occur to an assembled inboard CV joint if it is overplunged outward from the joint housing. Never use a hammer to remove or install the halfshafts from the hub. Never use the halfshaft assembly as a lever to position other components. Always support the free end of the halfshaft. Do not allow the boots to contact sharp edges or hot exhaust components. Handle the halfshaft only by the interconnecting shaft to avoid pull-apart and potential damage to the CV joints. Excessive pulling force on the interconnecting shaft between joints of the halfshaft will result in internal joint damage. Axial loads used in assisting removal must be applied through the inboard joint housing only. Do not drop assembled halfshafts. The impact will cut the boots from the inside without evidence of external damage. Do not remove the outer CV joint by pulling on the interconnecting shaft. Inspect all machined surfaces and splines for damage.

DIAGNOSTIC TESTS

REAR DRIVE HALFSHAFTS

Refer to **DRIVELINE SYSTEM - GENERAL INFORMATION** article.

REMOVAL AND INSTALLATION

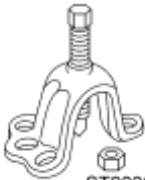
HALFSHAFT

Special Tools

Illustration	Tool Name	Tool Number
	Axle Seal Protector	205-816

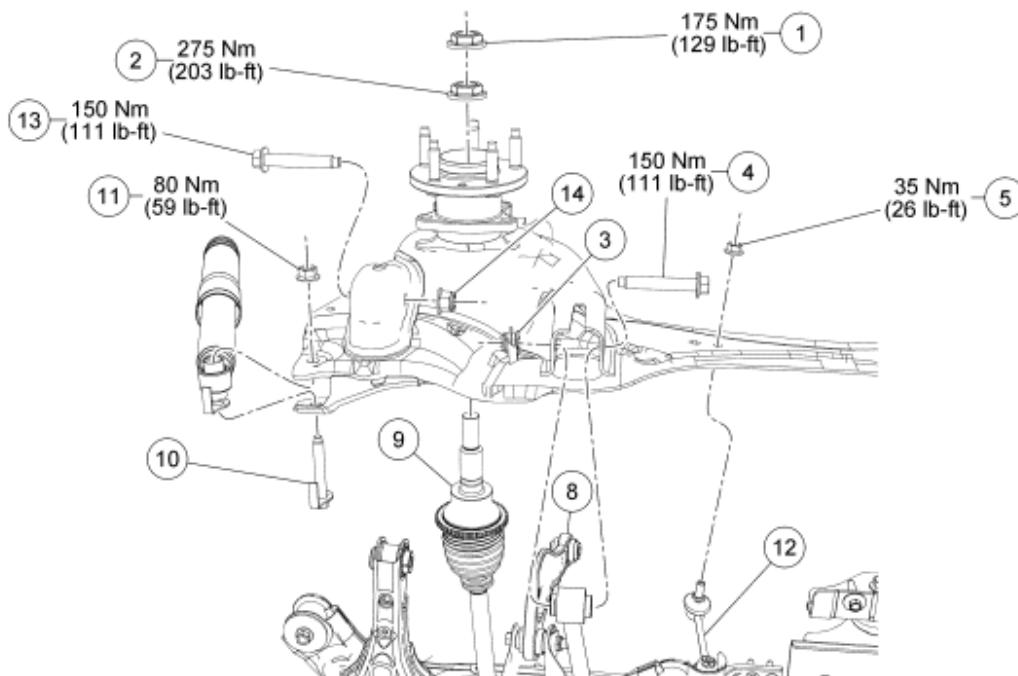
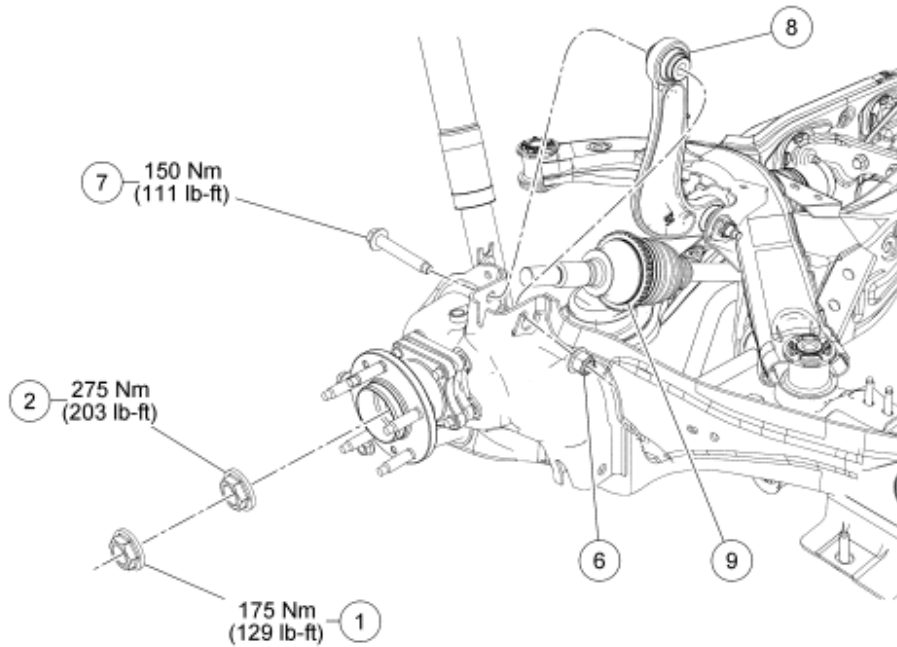
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 ST2828-A		
 ST2330-A	Front Hub Remover	205-D070 (D93P-1175-B) or equivalent

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N0076121

Fig. 1: Exploded View Of Halfshaft Components With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W712435	Outer hub bearing nut
2	W712435	Inner hub bearing nut

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3	W520516	Toe link outboard nut
4	W500766	Toe link outboard bolt
5	W520414	Stabilizer bar link nut
6	W520516	Upper arm outboard nut
7	W500766	Upper arm outboard bolt
8	5500	Upper control arm
9	4B402	Halfshaft assembly
10	W712459	Lower shock absorber flag bolt
11	W520214	Lower shock absorber nut
12	5A486	Stabilizer bar link
13	W500766	Lower arm outboard bolt
14	W520516	Lower arm outboard nut

REMOVAL

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.
2. Remove the rear wheel speed sensor. For additional information, refer to **VEHICLE DYNAMIC SYSTEMS** article.
3. Remove and discard the inner and outer rear wheel hub nuts.
4. Remove the brake caliper hose bracket bolt.

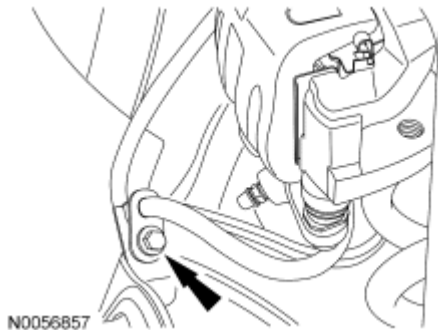


Fig. 2: Locating Brake Caliper Hose Bracket Bolt
Courtesy of FORD MOTOR CO.

5. Remove the rear brake disc. For additional information, refer to **REAR DISC BRAKE** article.
6. Remove the upper arm outboard bolt.

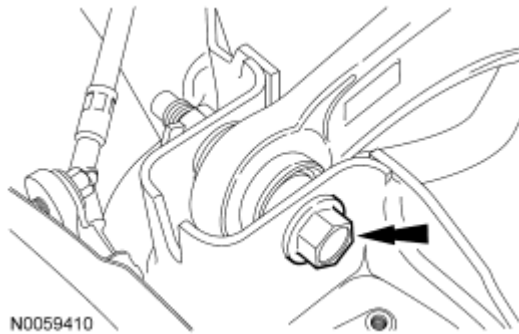


Fig. 3: Locating Upper Arm Bolt
Courtesy of FORD MOTOR CO.

7. Lift the upper arm from the knuckle.

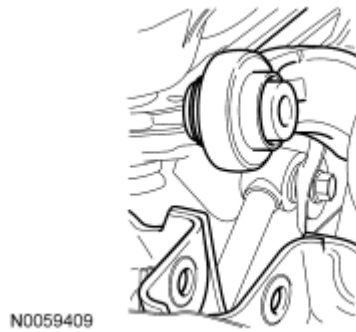


Fig. 4: Lifting Upper Arm From Knuckle
Courtesy of FORD MOTOR CO.

8. Support the lower arm and remove the lower shock nut and bolt.

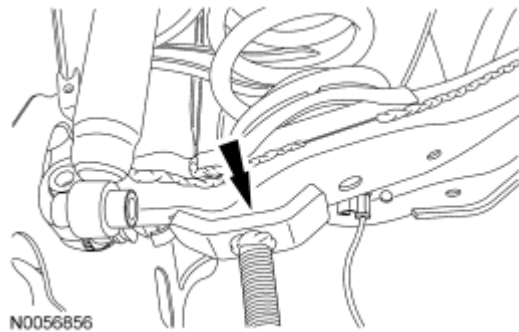


Fig. 5: Supporting Lower Arm
Courtesy of FORD MOTOR CO.

9. Remove the stabilizer link upper nut.

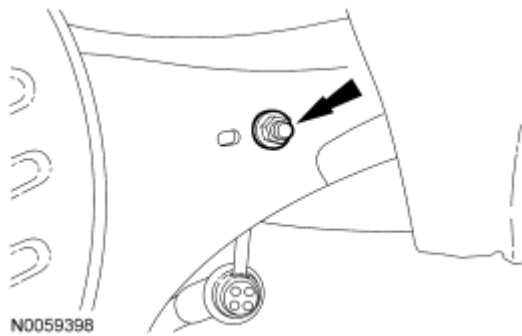


Fig. 6: Locating Stabilizer Link Upper Nut
Courtesy of FORD MOTOR CO.

10. Remove the outer toe link nut and bolt.

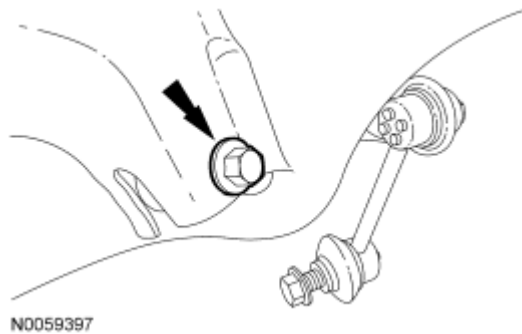


Fig. 7: Locating Outer Toe Link Nut & Bolt
Courtesy of FORD MOTOR CO.

11. Support the wheel knuckle and remove the lower arm nut and bolt.

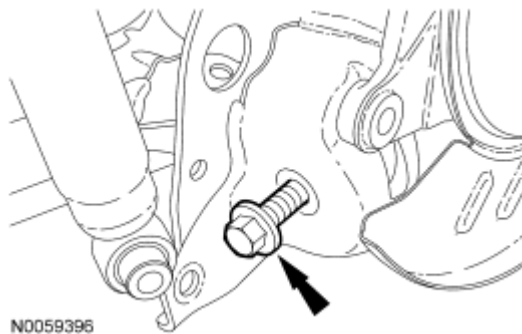
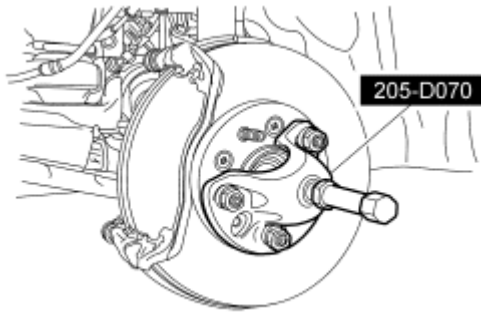


Fig. 8: Locating Lower Arm Bolt
Courtesy of FORD MOTOR CO.

12. Using the special tool, separate the halfshaft from the rear axle hub assembly.

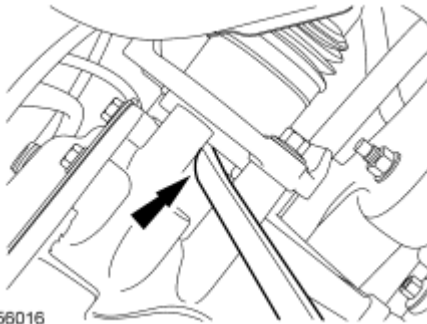


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Fig. 9: Separating Halfshaft From Wheel Hub Using Special Tool (205-D070)
Courtesy of FORD MOTOR CO.

CAUTION: Do not damage the oil seal when removing the axle halfshaft from the differential.

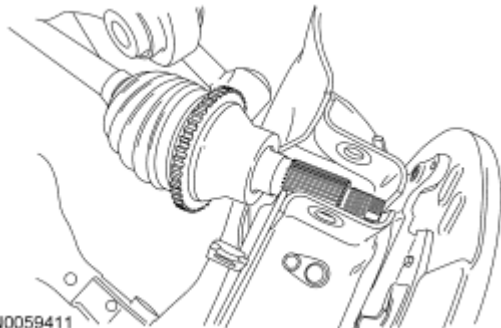
13. Using a suitable pry bar, remove the halfshaft.



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Fig. 10: Removing Halfshaft Using A Suitable Pry Bar
Courtesy of FORD MOTOR CO.

14. Position the halfshaft up through the wheel knuckle opening and remove the halfshaft.



N0059411

Fig. 11: Identifying Halfshaft & Wheel Knuckle
Courtesy of FORD MOTOR CO.

15. Remove and discard the circlip from the stub shaft.

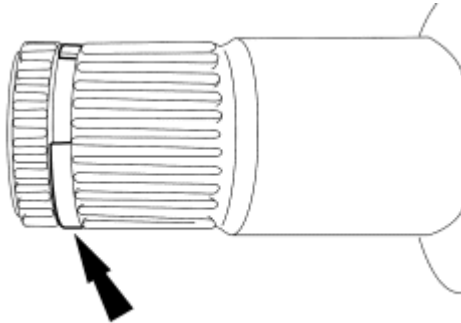


Fig. 12: Locating Halfshaft Circlip
Courtesy of FORD MOTOR CO.

INSTALLATION

1. Install a new 28 mm (1.102 in) circlip on the stub shaft.

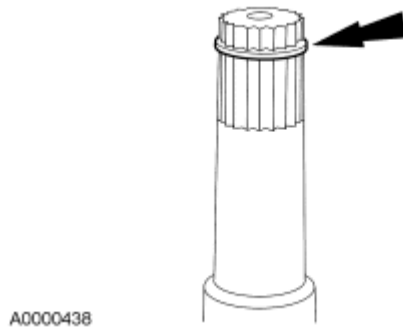


Fig. 13: Locating Intermediate Shaft Circlip
Courtesy of FORD MOTOR CO.

NOTE: Make sure the oil seal protector is correctly aligned with the differential oil seal during installation.

2. Using the special tool, install the halfshaft in the differential.

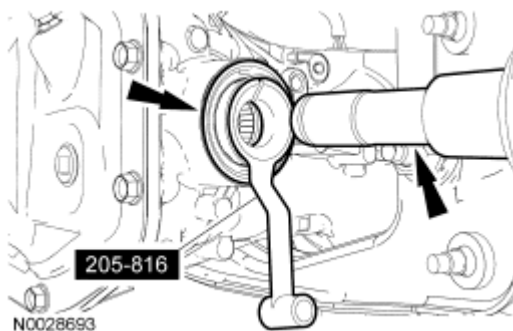


Fig. 14: Installing Halfshaft In Differential Using Special Tool (205-816)
Courtesy of FORD MOTOR CO.

3. Install the stub shaft in the rear drive unit.
 - Make sure the circlip locks in the side gear.

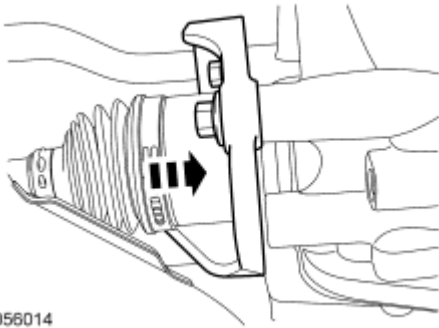


Fig. 15: Installing Stub Shaft In Rear Drive Unit
Courtesy of FORD MOTOR CO.

4. Slide the outboard CV joint down through the knuckle.

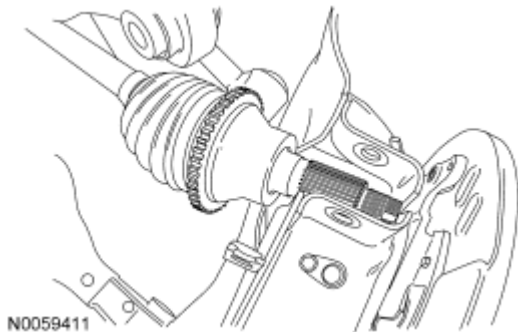


Fig. 16: Identifying Halfshaft & Wheel Knuckle
Courtesy of FORD MOTOR CO.

5. Position the halfshaft outer CV joint through the hub bearing.
6. Using a suitable tool, install the outer halfshaft end into the hub assembly.

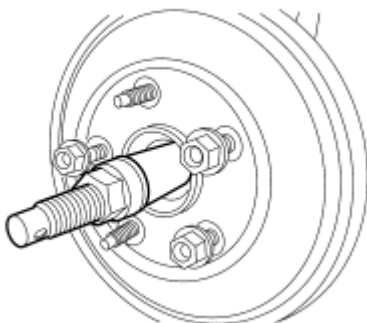


Fig. 17: Locating Halfshaft On Wheel Hub
Courtesy of FORD MOTOR CO.

7. Position the upper arm outboard end and install the bolt.
 - Tighten to 150 Nm (111 lb-ft).

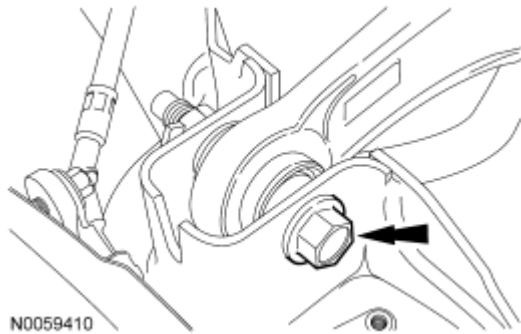


Fig. 18: Locating Upper Arm Bolt
Courtesy of FORD MOTOR CO.

8. Position the lower arm and install the lower arm bolt and nut.
 - Tighten to 150 Nm (111 lb-ft).

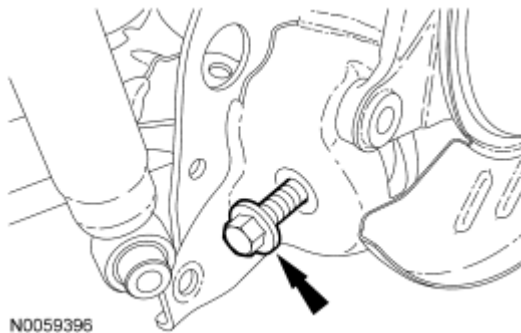


Fig. 19: Locating Lower Arm Bolt
Courtesy of FORD MOTOR CO.

CAUTION: Do not tighten the rear wheel hub nut with the vehicle on the ground. The nut must be tightened to specification before the vehicle is lowered to the ground. Wheel bearing damage will occur if the wheel bearing is loaded with the weight of the vehicle applied.

9. Install the 2 new rear wheel hub nuts.
 - Tighten the inner hub nut to 275 Nm (203 lb-ft).
 - Tighten the outer hub nut to 175 Nm (129 lb-ft).
10. Install the rear toe link bolt.
 - Tighten to 200 Nm (148 lb-ft).

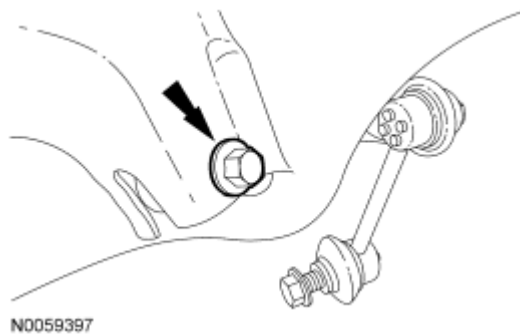


Fig. 20: Locating Outer Toe Link Nut & Bolt
Courtesy of FORD MOTOR CO.

11. Install stabilizer link upper nut.
 - Tighten to 35 Nm (26 lb-ft).

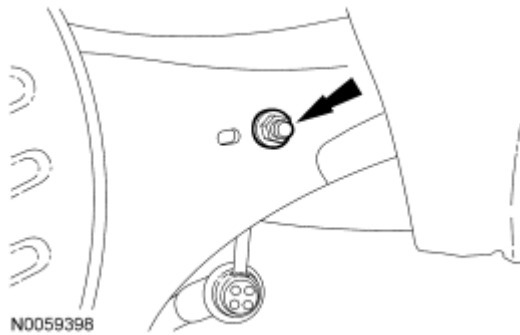


Fig. 21: Locating Stabilizer Link Upper Nut
Courtesy of FORD MOTOR CO.

12. Install the lower shock absorber bolt.
 - Tighten to 115 Nm (85 lb-ft).
13. Install the rear wheel speed sensor. For additional information, refer to **VEHICLE DYNAMIC SYSTEMS** article.
14. Install the rear brake disc. For additional information, refer to **REAR DISC BRAKE** article.
15. Install the brake caliper hose bracket bolt.
 - Tighten to 15 Nm (11 lb-ft).

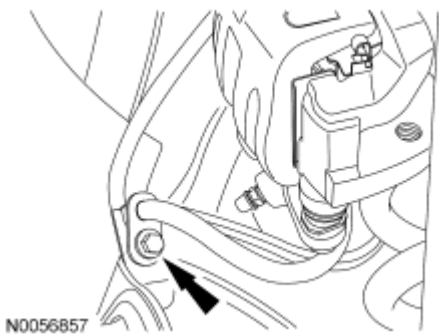


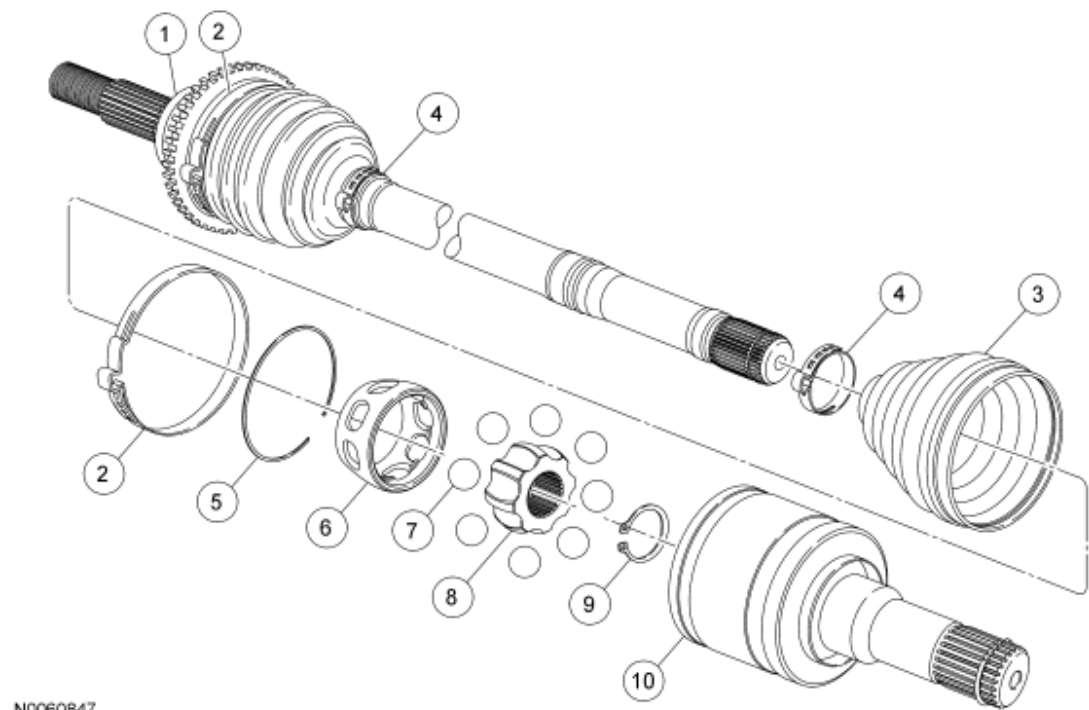
Fig. 22: Locating Brake Caliper Hose Bracket Bolt
Courtesy of FORD MOTOR CO.

16. Install the tire and wheel. For additional information, refer to **WHEELS AND TIRES** article.

CONSTANT VELOCITY (CV) JOINT BOOT

Special Tools

Illustration	Tool Name	Tool Number
A line drawing of a metal clamp with two bolts. The part number ST1301-A is written below the drawing.	Installer, Constant Velocity Joint Boot Clamp	205-343 (T95P-3514-A)



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Fig. 23: Exploded View Of Constant Velocity (CV) Joint Boot
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	4B202	Outboard constant velocity (CV) joint and interconnecting shaft assembly
2	4B478	Boot clamps (part of 4L537)
3	3L537	CV joint boot (part of 4L537)
4	4B478	Boot clamps (part of 4L537)
5	4B422	Retaining ring
6	-	Ball cage (part of 3C092)
7	-	Ball (8 required) (part of 3C092)
8	-	Inner race (part of 3C092)
9	3B426	Snap ring
10	3C092	Inboard CV joint housing

DISASSEMBLY

1. Remove the halfshaft assembly. For additional information, refer to **Halfshaft** .
2. Remove and discard the inboard constant velocity (CV) joint boot clamps.



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Fig. 24: Identifying Inboard Constant Velocity (CV) Joint Boot Clamps
Courtesy of FORD MOTOR CO.

3. With the inboard CV joint boot removed from the housing, remove the retaining ring.

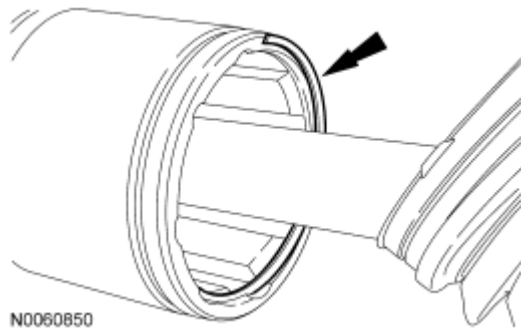


Fig. 25: Locating Retaining Ring
Courtesy of FORD MOTOR CO.

4. Remove the 8 balls and outer race.

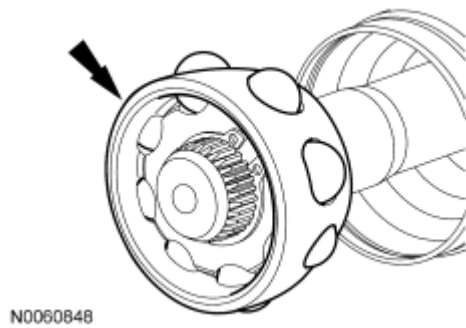


Fig. 26: Locating Outer Race Balls
Courtesy of FORD MOTOR CO.

5. Remove the CV joint housing snap ring.

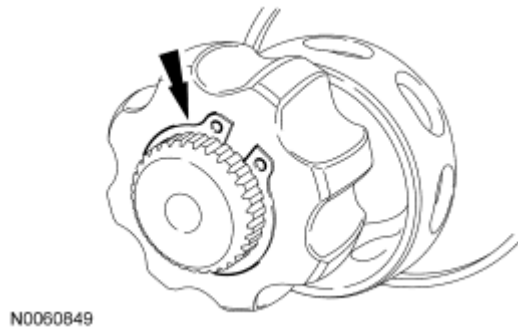
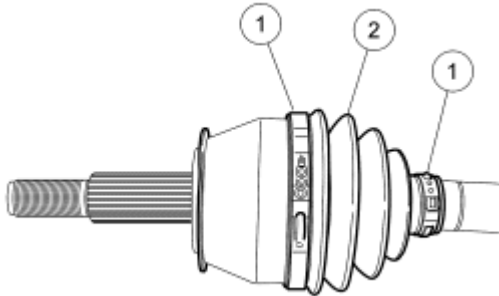


Fig. 27: Locating Retainer Snap Ring
Courtesy of FORD MOTOR CO.

6. Remove and discard the boot.

NOTE: The outboard CV joint is not removable from the halfshaft. The boot must be removed or installed from the inboard CV joint side of the shaft.

7. For the outboard CV joint boot, carry out the following:
 1. Remove and discard the boot clamps.
 2. Remove and discard the boot.

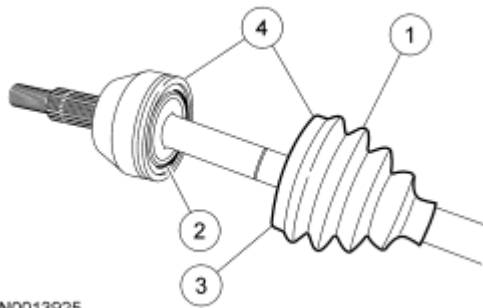


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Fig. 28: Identifying Boot Clamps And Boot
Courtesy of FORD MOTOR CO.

ASSEMBLY

1. For the outboard CV joint, carry out the following:
 1. Slide the boot on the interconnecting shaft.
 2. Pack the outboard CV joint with 80 g (2.82 oz) of grease supplied in the boot kit.
 3. Spread any remaining grease evenly inside the boot.
 4. Install the boot by seating it in the groove in the CV joint housing.



N0013925

Fig. 29: Identifying CV Joint Housing
Courtesy of FORD MOTOR CO.

2. Using the special tool, install the 79.6 mm (3.13 in) boot clamps on the large end of the boot.

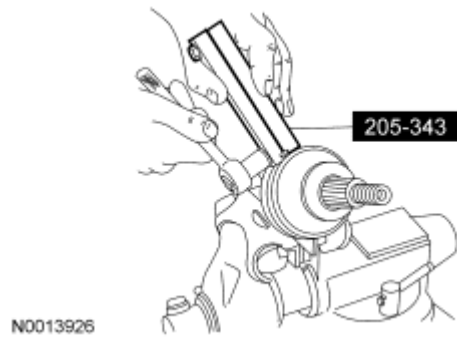


Fig. 30: Installing Boot Clamps Using Special Tool (205-343)
Courtesy of FORD MOTOR CO.

3. Install the small clamp on the small end of the boot.

CAUTION: Install a new circlip every time the halfshaft is removed from the axle or damage to the component may occur.

CAUTION: Start one end of the circlip in the groove and work the circlip over the halfshaft and into the groove to prevent the circlip from overexpanding or damage to the clamp may occur.

4. Install a new retainer circlip.

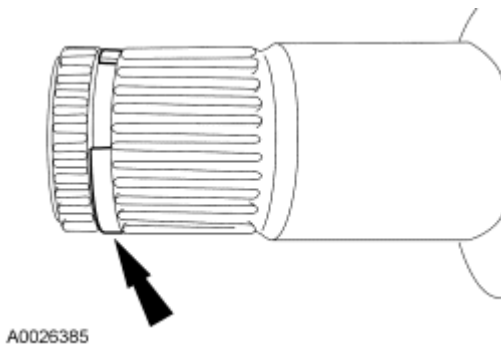
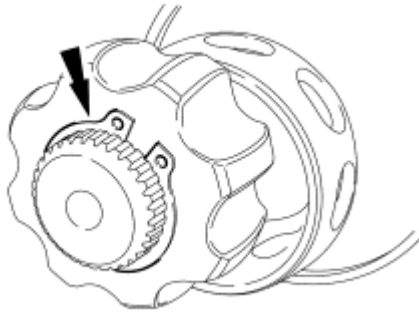


Fig. 31: Locating Halfshaft Circlip
Courtesy of FORD MOTOR CO.

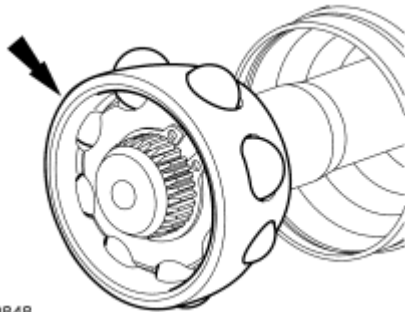
5. Position the small clamp on the interconnecting shaft before installing the boot.
6. Position the boot on the interconnecting shaft.
7. Install the inner race and the retainer snap ring.



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Fig. 32: Locating Retainer Snap Ring
Courtesy of FORD MOTOR CO.

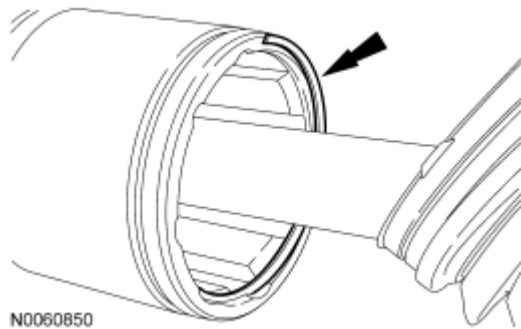
8. Install the outer race with 8 balls.



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Fig. 33: Locating Outer Race Balls
Courtesy of FORD MOTOR CO.

9. Insert the assembly in the housing.
10. Install the retaining ring.



N0060850

Fig. 34: Locating Retaining Ring
Courtesy of FORD MOTOR CO.

11. For the inboard CV joint, carry out the following steps in sequence:
 1. Pack the inboard CV joint housing with 108 g (3.80 oz) of grease supplied in the boot kit.

2. Spread any remaining grease evenly inside the boot and CV joint.
3. Install the inboard CV joint housing, seating the boot in the groove in the housing.
12. Set the halfshaft assembled length to specifications. For additional information, refer to **SPECIFICATIONS** .
 1. Measure the entire assembly length.
 2. Push in or pull out on the inner joint as necessary to adjust the halfshaft assembled length.
 3. Hold the inner joint to prevent the assembled length from changing and insert a soft, flat tool between the boot and the joint to equalize the pressure.

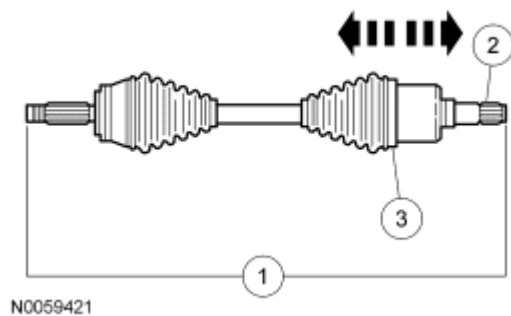


Fig. 35: Pushing In Or Pulling Out On Inner Joint As Necessary To Adjust Halfshaft Assembled Length

Courtesy of FORD MOTOR CO.

13. Using the special tool, install the 77.5 mm (3.05 in) boot clamps on the large end of the boot.

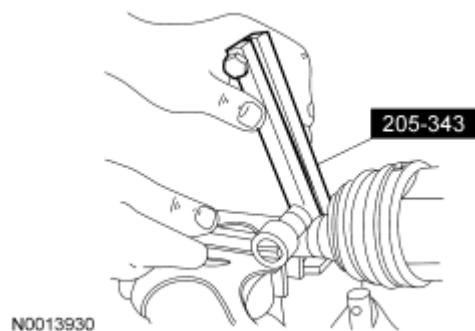


Fig. 36: Installing Both Boot Clamps Using Special Tool (205-343)

Courtesy of FORD MOTOR CO.

14. Install the smaller boot clamp.
15. Install the halfshaft assembly. For additional information, refer to **Halfshaft** .